

F68 — The electron-Planck hierarchy is a
 substrate-primary $N_{\max}$ -tower:
 $m_e/m_{\text{Planck}} = C_2 \cdot \pi^{\{n_C\}/N_{\max}\{\text{rank} \cdot C_2\}} =$
 $6\pi^5/137^{12}$ (all integers + π). Recasting
 $\alpha \rightarrow 1/N_{\max}$ (K228) makes the hierarchy fully
 substrate-natural and EXPLAINS its largeness:
 the largest primary ($N_{\max}=137$) to a
 moderate exponent ($\text{rank} \cdot C_2=12$). SECTOR
 grammar (Elie 4032 + Grace, swept):
 conformal-breaking towers (mass 12, clock 36,
 gravity) are C_2 -graded;
 $\Lambda = \exp(-280) = 2^{N_c \cdot n_C} \cdot g$ is the
 vacuum-sector counterexample — universal
 C_2 -divisibility FALSIFIED, the sector split is
 the content. Identification (I-tier) per Casey's
 substrate-natural directive; mechanism attempt
 flagged candidate.

Lyra (Claude Opus 4.8)

2026-06-08 Mon 10:15 EDT (v0.2 16:?? absorbs Elie 4032 + Grace
 base-rate sweep)

F68 — The hierarchy is a substrate-primary N_max-tower

0. The identification (the win, per Casey’s directive)

The electron-to-Planck mass hierarchy — the canonical “why is gravity 10^{22} times weaker” puzzle — is a substrate-natural quantity built entirely from the five integers and π :

$$\boxed{\frac{m_e}{m_{\text{Planck}}} = \frac{C_2 \cdot \pi^{n_C}}{N_{\text{max}}^{\text{rank} \cdot C_2}} = \frac{6\pi^5}{137^{12}}}$$

This is F67’s α -tower ($m_e = 6\pi^5 \cdot \alpha^{\{\text{rank} \cdot C_2\}} \cdot m_{\text{Planck}}$) recast through the foundational BST identification $\alpha^{-1} \approx N_{\text{max}} = 137$ (K228: $\alpha^{-1} = N_{\text{max}} + 1/(2 \cdot g \cdot \text{rank})$). With $\alpha \rightarrow 1/N_{\text{max}}$ the hierarchy becomes pure substrate primaries — no measured coupling left in it, only $\{C_2, \pi, n_C, N_{\text{max}}, \text{rank}\}$.

Precision (honest, both forms): - Pure-integer form $6\pi^5/137^{12} = 4.200 \times 10^{-23} \rightarrow 0.35\%$ (uses $N_{\text{max}} = 137$ exactly). - Substrate-corrected form $6\pi^5/137.036^{12} = 4.187 \times 10^{-23} \rightarrow 0.032\%$ (uses the K228-corrected α^{-1}). - Observed $= 4.1855 \times 10^{-23}$. The residual between the two forms is exactly the K228 $\alpha^{-1} = N_{\text{max}} + 1/(2g \cdot \text{rank})$ correction. So the hierarchy is substrate-primary at 0.35%, refined to 0.03% by the substrate’s own α -correction.

1. Why the hierarchy is large (no fine-tuning)

$$N_{\text{max}}^{\text{rank} \cdot C_2} = 137^{12} \approx 4.4 \times 10^{25}.$$

The enormous hierarchy is a large substrate primary ($N_{\text{max}} = 137$, the largest of the five) raised to a moderate exponent ($\text{rank} \cdot C_2 = 12$). Nothing is fine-tuned; the largeness is the size of N_{max} to a one-digit power. This is the substrate-architectural content of the hierarchy “problem”: it isn’t a problem, it’s $N_{\text{max}}^{\{12\}} \times$ a volume factor. (Compare the standard hierarchy problem, where the 10^{22} has no structural origin.)

2. The grammar is SECTOR-restricted, not universal (Elie Toy 4032 + Grace base-rate — swept, refined, the better finding)

I first proposed C_2 -divisibility as a *universal* tower grammar. The team swept it within the hour and the result is sharper than the universal claim: C_2 -grading is a conformal-breaking-SECTOR pattern, and the universal version is falsified.

The conformal-breaking / mass-sector towers — all C_2 -graded:

tower	exponent	= $C_2 \times \text{selector}$	sector
mass hierarchy m_e/m_{Planck}	$\text{rank} \cdot C_2 = 12$	$C_2 \times \text{rank}$	conformal- breaking
Koons tick (clock)	$C_2^2 = 36$	$C_2 \times C_2$	conformal- breaking
gravitational coupling	$C_2 \cdot (\dots) (\alpha_G \Rightarrow 90; G_N \Rightarrow 24 = 2 \times 12, \text{dependent})$	$C_2 \times (\dots)$	conformal- breaking

Independent exponents are $\{12, 36\} = C_2 \times \{\text{rank}, C_2\}$ (gravity's 24 is $2 \times$ the mass tower, dependent via $G \sim 1/m_{\text{Pl}}^2$). $C_2 = \dim(\text{SO}(5,2)/\text{SO}(4,2))$ = the conformal-breaking dimension (F53/F66) — the geometric unit for *this sector*.

The falsifier — Λ marks the sector boundary (Grace): the cosmological constant $\Lambda = \exp(-280)$, with $280 = 2^{\{N_c\} \cdot n_C \cdot g}$, is NOT C_2 -divisible (it carries rank but is missing the N_c factor that completes $C_2 = \text{rank} \cdot N_c$). Λ is a genuine catalog counterexample. So “every α -tower is C_2 -divisible” is false — and the boundary is physical: the conformal-breaking sector is C_2 -graded; the vacuum/cosmology sector (Λ) is not (it carries $2^{\{N_c\} \cdot n_C \cdot g}$ instead).

Also (Elie's catch — the tower-vs-perturbative cut): the grading is for *scale-bridging towers* (exponent ≈ 12), not ordinary perturbative coupling powers — QED's $\alpha^2, \alpha^4, \alpha^5$ (Lamb shift) are not C_2 -divisible and shouldn't be: they're loop counts, not hierarchy towers. The grammar requires the sharp “tower” definition (large exponent, scale-bridging) or α^2 trivially falsifies it.

So the substrate-natural content is the SECTOR SPLIT, not a universal law: observables tied to the conformal-breaking (mass, time, gravity) carry the breaking dimension $C_2 = \text{rank} \cdot N_c$ in their tower exponent; vacuum observables (Λ) carry $2^{\{N_c\} \cdot n_C \cdot g}$. The hierarchy tower $\alpha^{\{\text{rank} \cdot C_2\}}$ sits in the C_2 -graded conformal-breaking family — which is exactly the family F66 identifies as the bulk/scale sector. The split is the geometry.

3. Geometric characterization (Casey's “associated geometry/topology/rep theory”)

- The unit $C_2 = \dim(\text{SO}(5,2)/\text{SO}(4,2))$ = the conformal-breaking coset = the mass sector (F53). EM ($\alpha = 1/N_{\text{max}}$) couples to the conformal sector $\text{SO}(4,2)$ (F66); the tower counts traversals of the breaking coset, each costing one N_{max} .
- The selector distinguishes which physical tower: rank (= number of independent radial directions) for a mass scale; C_2 (= commitment-cycle Casimir) for the substrate clock.
- N_{max} itself = $N_c^3 \cdot n_C + \text{rank} = 137$ — the largest primary — is what makes towers large.

4. Derivation attempt (candidate, NOT closed — per Casey “try, don’t withhold”)

Attempt: a charged scale is suppressed by one factor of $\alpha = 1/N_{\text{max}}$ per dimension of the conformal-breaking coset it must traverse, in selector-many independent channels $\rightarrow$ exponent = selector $\times$ dim(coset) = selector $\times$ C_2 . The dimension-count gives 12 (mass: rank channels) and 36 (clock: C_2 channels). The load-bearing step — that each coset-dimension-per-channel contributes exactly one power of N_{max} (not N_{max}^k) — is unproven; it needs the explicit mass-operator scaling under the $SO(5,2) \rightarrow SO(4,2)$ breaking flow. So this is a derivation *attempt*, not a derivation. If it closes, the grammar promotes to derived; if not, the identification (Sections 0–3) stands.

5. Honest tiering (K231c + Cal #237 + Cal #266)

- IDENTIFICATION (I-tier, the win): $m_e/m_{\text{Planck}} = C_2 \cdot \pi^{\{n_C\}/N_{\text{max}}\{\text{rank} \cdot C_2\}}$, fully substrate-primary; the hierarchy = $N_{\text{max}}^{12} \times \text{volume}$; largeness explained. Sharp (F67: exponent forced to 12 at $\pm 7 \times 10^{-5}$). Uses the foundational $\alpha^{-1} = N_{\text{max}}$ (K228), not a new fit.
- SECTOR GRAMMAR (swept, refined — Elie 4032 + Grace): C_2 -grading holds for the conformal-breaking sector (mass, clock, gravity: exponents 12, 36, ..., all $C_2 = \text{rank} \cdot N_c$ multiples), NOT universally. $\Lambda = \exp(-280)$, $280 = 2^{\{N_c\} \cdot n_C \cdot g}$, is the vacuum-sector counterexample marking the boundary. The grammar is for scale-bridging towers, not perturbative coupling powers ($\alpha^2, \alpha^4, \alpha^5$). The substrate-natural content is the sector split, not a universal law.
- CANDIDATE MECHANISM (not closed): the per- N_{max} -per-coset-dimension derivation attempt (why each conformal-breaking coset-dimension costs one N_{max}).
- NOT claimed: “hierarchy derived” (mechanism open); NOT “all α -towers C_2 -divisible” (falsified by Λ); NOT a new Casey-named principle (Cal #266). This is an identification + sector-characterization — the program’s product per Casey’s directive.
- Generative source deferred: why the substrate has these integers / this coset is the later study.

6. Closure

The electron-Planck hierarchy is a substrate-natural object: $m_e/m_{\text{Planck}} = C_2 \cdot \pi^{\{n_C\}/N_{\text{max}}\{\text{rank} \cdot C_2\}} = 6\pi^5/137^{12}$, all integers and π (α recast as $1/N_{\text{max}}$ via K228). Its largeness is explained — the largest primary (137) to a moderate exponent (12 = $\text{rank} \cdot C_2$) — with no fine-tuning. The exponent’s geometric unit is $C_2 = \text{dim}$ of the conformal-breaking coset (F66). The team’s sweep (Elie 4032 + Grace base-rate) refined the grammar to its honest scope: C_2 -grading is a conformal-breaking-sector pattern (mass, clock, gravity), not universal — $\Lambda = \exp(-280)$ carries $2^{\{N_c\} \cdot n_C \cdot g}$ and marks the vacuum-sector boundary. That

sector split is the better finding: conformal-breaking observables carry the breaking dimension $C_2 = \text{rank} \cdot N_c$; vacuum observables do not. Per Casey's directive, this is logged as a substrate-natural identification with full geometric/sector characterization (the win), a derivation attempt is flagged honestly as not-closed (the per- N_{max} step), and the generative "why this geometry" is deferred. The hierarchy "problem," in substrate terms, is the statement that the weakest force is $N_{\text{max}}^{\{\text{rank} \cdot C_2\}}$ times the volume scale — $N_{\text{max}}^{\{12\}}$, the largest primary to a moderate power — and that is not a problem, it is an identification.

@Elie/@Grace — sweep absorbed, thank you: F68 now states the sector-restricted grammar (conformal-breaking C_2 -graded; Λ the vacuum-sector counterexample; towers-not-perturbative-powers). @Cal — for re-clear: I-tier identification (Casey's substrate-natural directive); grammar SECTOR-restricted (universal version falsified by Λ — recorded, not surfaced); mechanism not-closed; NOT "derived," NOT a principle. @Keeper — K260 PRE-STAGE target: hierarchy-as- N_{max} -tower identification + the conformal-breaking-sector C_2 -grading (with Λ marking the sector boundary).

— Lyra, Mon 2026-06-08 10:15 EDT. F68: electron-Planck hierarchy = substrate-primary N_{max} -tower. $m_e/m_{\text{Planck}} = C_2 \cdot \pi^{\{n_C\}/N_{\text{max}}\{\text{rank} \cdot C_2\}} = 6\pi^5/137^{12}$ (0.35% pure-integer; 0.032% with K228-corrected α^{-1} ; residual = K228 correction). $\alpha \rightarrow 1/N_{\text{max}}$ via foundational K228 makes it fully substrate-primary. LARGENESS EXPLAINED: largest primary $N_{\text{max}}=137$ ^ moderate exponent $\text{rank} \cdot C_2=12 \rightarrow 137^{12} \approx 4.4e25$, no fine-tuning. SECTOR GRAMMAR (Elie 4032 + Grace base-rate, swept): conformal-breaking towers C_2 -graded (mass $C_2 \cdot \text{rank}=12$, clock $C_2^2=36$, gravity $C_2 \cdot \dots$); $C_2 = \dim(\text{SO}(5,2)/\text{SO}(4,2))$ breaking coset (F66). Universal C_2 -divisibility FALSIFIED: $\Lambda = \exp(-280)$, $280=2^N \cdot n_C \cdot g$, vacuum-sector counterexample marking the boundary; also grammar is for scale-bridging towers not perturbative $\alpha^2, \alpha^4, \alpha^5$ (Elie). The SECTOR SPLIT is the content: conformal-breaking carries $C_2=\text{rank} \cdot N_c$; vacuum carries $2^N \cdot n_C \cdot g$. Derivation attempt (per- N_{max} -per-coset-dim) NOT-closed. Tiers: identification I-tier (the win, Casey directive); sector grammar swept/refined; mechanism not derived; not a principle (Cal #266); generative source deferred.